

What Affects Digital Divide for the Elderly During the Pandemic: An Empirical Study of Use of COVID-19 E-government Services

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With focus on the deepening digital divide among the elderly after the COVID-19 pandemic, this study aims to empirically identify the multidimensional factors affecting the digital divide. As for the research sample, survey data of 428 elderly individuals aged 55 and over from South Korea's 'National report on digital divide 2020' were analyzed. As for the research model, the Bélanger and Carter's model was modified and applied to the South Korean context. The Independent variables included demographic factors such as household type, income, education, age and gender, as well as digital capacity with PC and mobile devices, while the dependent variable measured the level of e-government service use. ANOVA and crosstab analysis were hired for the analysis. The analysis revealed that all demographic factors namely income, education level, age and gender but household type had a statistically significant effect on the elderly's level of e-government service use. Furthermore, digital capacity, encompassing both PC and mobile device proficiency, was also proven to be a critically important predictor of service utilization. This study confirms that the digital divide among the elderly is still influenced by key demographic characteristics. In conclusion, for the successful digital inclusion of the elderly, policy support for education that promotes practical digital skills is required, in addition to considering structural factors.

Key words: digital divide, digital literacy, e-government, COVID-19

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I. Introduction

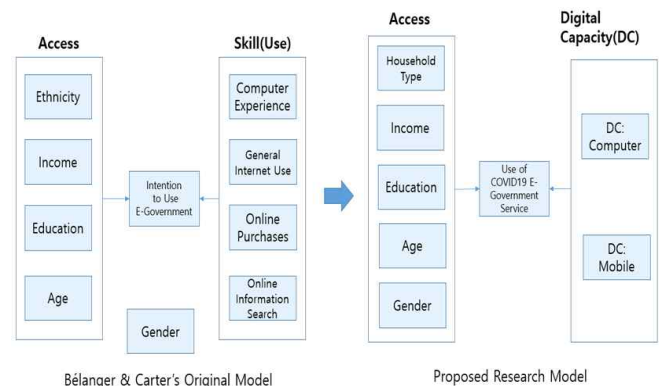
Since the COVID-19 pandemic swept all over the world, its long-term impact has accelerated the digital transformation of public and private services. To this end, video telephony, e-commerce and working remotely have become common. For example, ZOOM, a global online meeting platform, experienced exponential user growth during the pandemic. E-commerce platforms have also shown remarkable growth. Notably, e-government platforms have become more accessible and convenient. However, such drastic growth of those online platforms may exacerbate the digital divide among the socially marginal population. The COVID-19 pandemic has significantly deepened and accelerated the digital divide (Beaunoyer, Dupéré and Guittou 2020). Digital divide is particularly pronounced among the elderly, who tend to have a lower level of access to the Internet and of proficiency in using digital technologies. As a result, they often face challenges in obtaining information, accessing healthcare services and maintaining social connections (Robinson, Schulz, Khilnani, Ono, Cotten and Duque 2020). Against this backdrop, a systematic investigation of the drivers that either exacerbate or mitigate the digital divide among the elderly is urgently required. Accordingly, this study seeks to identify influential factors regarding the digital divide among the elderly.

The sample data were obtained from South Korea's national report on digital divide (Ministry of Science and ICT 2021). This report has been published annually. For the year 2020, it contains a questionnaire on the COVID-19 e-government services. Against the COVID-19 pandemic, the Korean government introduced new e-government services including a real-time COVID-19 bulletin board and an online registration of financial support package of COVID-19 vaccination appointment. Since then, the Korean society has gradually adapted to a post-pandemic normal through the widespread digital service adoption, resulting in a rapid change in life patterns from offline to online. The data show how the COVID-19 e-government services has changed the digital environment. Thus we have decided to use them as the research sample and investigated what would facilitate the digital divides among the elderly.

II. Background Literature

1. Digital Divide

We employ the Bélanger and Carter framework (2009) to investigate the effect of digital divide on e-government usage. However, we have made some modifications to their framework that defines digital divide in terms of access and skill and proposed a new model which also considers the social context of the COVID-19 driven digitization. The purpose of these modifications is to enhance the model's analytical utility and to adapt the framework to the unique context of accelerated digitization during the COVID-19 pandemic in South Korea. In the proposed model, we made the following modifications: we replaced (1) ethnicity from the Bélanger and Carter (2009) framework with household type and gender; (2) skill set with digital capacity and (3) changed the dependent variable from 'intention to use e-government' to "use of COVID-19 e-government services". The proposed model is shown in <Figure 1>.



<Figure 1> Bélanger & Carter's Original Model and Proposed Research Model

2. Access Divide

The literature posits that low access to the Internet creates digital divide (Hoffman, Novak and Schlosser 2000; Mossberger, Tolbert and Stansbury 2003). Importantly, the digital divide extends beyond issues of basic access to encompass disparities in digital skills and the ability to derive meaningful benefits from technology use (Scheerder, van Deursen and van Dijk 2017). Van Dijk (2017) theorizes that unequal access to digital-related resources creates disparities in the use of technology. He and other scholars also theorize that demographic factors such as behavior and social status contribute to unequal Internet access (Pick and Sarkar 2016; Van Dijk 2017). Furthermore, Van Deursen and Van Dijk (2010) emphasize that digital divide is no longer simply about access, but about the possession of internet skills. While these skills are essential for their engagement with digital platforms such as e-government services, they are often unequally distributed and used among social groups. As a result, the absence of these skills has the elderly excluded from opportunities for digital proficiency.

Moreover, digital divide is not uniform within the older population. Empirical studies have shown that factors such as age, educational attainment, income and household composition significantly influence the degree of digital exclusion experienced by the elderly (Friemel 2016). Based on both Van Dijk's theoretical assertions and the Bélanger and Carter's framework, we developed five independent variables on access to the Internet: household type, income, education, age and gender. Notably, we replaced ethnicity from the Bélanger and Carter framework with household type and gender. This is because the research data were collected from the Korean population, which is ethnically homogeneous. For this reason, it is pointless to set ethnicity in low diversity as a independent variable for our research model.

Some previous studies have indicated that persons around the elderly play a significant role in bridging digital divide (Harwood and McIntosh 2004; Peek, Luijckx, Rijnaard, Nieboer, van der Voort, Aarts and Wouters 2016; Kim, Choi and Lee 2022; Kim, 2024). They imply that seniors living with younger generations may have higher access to the Internet and are behaviorally inclined to use technology (Harwood and McIntosh 2004; Peek et al. 2016). Taipale (2013) classifies the research sample by the presence of a family partner and shows weak support for the effect of household type on the digital divide. This finding called for a need for further academic credit. Moreover, Van Dijk (2006) highlights the role of social support often embedded within household structure in facilitating digital skills acquisition. Thus, it is theoretically

plausible that household type affects not only access level but also skill development. Therefore, this study sets up a hypothesis tested against the effect of households on the digital divide to clarify it.

Unlike the original model, our proposed model investigates the effect of gender on digital divide. Bélanger and Carter (2009) argue that there are no disparities in the level of Internet use between genders anymore. However, we decided to include gender as an independent variable because other previous studies have indicated that gender may be significant (Van Slyke, Comunale and Belanger 2002). Moreover, since most of those over the age of 55 today belong to the first generation that experienced the Internet, it is necessary to confirm the effect of gender. Also Van Deursen and Van Dijk (2010) note that while overall gender differences in internet use have narrowed, women tend to happen to underestimate their digital competencies. Accordingly, performance gap in gender may exist in certain skill domains. This supports the rationale for examining gender as a potential explanatory factor in the digital divide among the elderly.

- H 1: Household type significantly facilitates the use of the COVID-19 e-government services.
- H 2: Income level significantly facilitates the use of the COVID-19 e-government services
- H 3: Education level significantly facilitates the use of the COVID-19 e-government services
- H 4: Age significantly facilitates the use of the COVID-19 e-government services
- H 5: Gender significantly facilitates the use of the COVID-19 e-government services

3. Digital Capacity Divide

In the proposed model, the skills side, one of the two main pillars for Bélanger and Carter's model, has been replaced with digital capacity for several reasons. First, the factors on the skill side in the Bélanger and Carter's model measure the level of individual IT skill (e.g., how well persons purchase items online or execute an online search). In contrast, we sought to determine the overall digital competency. This approach allowed us to perform a more comprehensive and holistic review of digital capacity as a whole. Second, the original model focuses on merely proficiency with PC. The original model is out-dated because for today mobile devices are increasingly being used for access to the Internet.. In this regard, we considered individuals' digital competency for both PC and mobile devices. A construct measuring digital capacity, rather than the IT skillset is, therefore, more suited for illustrating today's digital divide. According to Van Deursen and Van Dijk (2010), digital competence is not merely about technical execution. It should consider the ability to understand, interpret and utilize digital information effectively. For this reason, digital capacity in our model includes the expanded conception of digital skills including secure data management and strategic navigation of services. Also we substituted basic IT skills in the Bélanger and Carter research model with a broader digital capacity construct in our research model.

Existing literature consistently has demonstrated that the elderly possess a lower level of digital capacity than the younger generations (Kim et al. 2022; Kim 2024). The limited exposure to digital devices and the unfamiliarity with them have digitally marginalized the elderly (Harwood and McIntosh 2004; Peek et al. 2016). For example

they are away from social networking services and have trouble with using online financial transactions. As a result, the elderly often experience diminished confidence in their ability to critically assess digital information, safeguard personal data and navigate the complex interactive features that characterize many digital platforms (Choi 2021; Robinson et al. 2020). These challenges set substantial barriers for the elderly to attempt tasks that are relatively straightforward for younger people. (Kim et al. 2022; Kim 2024). Therefore, it is crucial to adopt digital capacity as a crucial component for our research model to comprehensively reflect the multifaceted nature of digital divide. In this regard, the questionnaire for our study measures digital competency with both PC and mobile devices. The questionnaire is quoted from a government survey measuring digital divide (Ministry of Science and ICT 2021). The reliabilities of the scale for digital proficiency with PC and mobile are .94 and .90, respectively.

H 6a: Level of digital capacity with PC significantly facilitates the use of the COVID-19 e-government services

H 6b: Level of digital capacity with mobile significantly facilitates the use of the COVID-19 e-government services

<Table 1> Construct of Digital Capacity

Questionnaire	
Computer	• I can install/remove/update necessary programs (software) on my computer.
	• I can connect my computer to wired or wireless internet and use it. (However, IP settings are excluded)
	• I can set up an environment in the web browser (Explorer, Chrome, Safari, etc.)
	• I can connect various external devices (digital camera, printer, scanner, portable hard drive, etc.) to my computer and use them.
	• I can transfer files from my computer to others over the Internet.
	• I can scan/repair malicious code (virus, spyware, etc.) on my computer.
	• I can create documents and materials using a computer (MS office suite, etc.)
Mobile	• I can configure the general environment setting including display, sound, security, alarm and input method, on my mobile device.
	• I can set up the wireless network (Wi-Fi) setting on my mobile device.
	• I can move files from my mobile device to my computer.
	• I can send files/photos from my mobile device to others.
	• I can install/remove/update necessary applications on my mobile device.
	• I can scan/repair malicious code on my mobile device (virus, spyware, etc.)
	• I can write documents or materials (memo, word, etc) on my mobile device.

4. Use of COVID-19 E-government Services

We changed the dependent variable from "intention to use e-government" in the original model to "use of COVID-19 e-government service." for our proposed model to reflect on the COVID-19 situation. The level of the use of the COVID-19 e-government services is the proxy indicator for measuring digital divide during the COVID-19 pandemic. Since the outbreak of COVID-19 in early 2020, there has been a major shift in the way citizens communicate with public institutions, with most government services being converted to digital platforms. This rapid digital transformation exert a significant impact on the elderly who have struggled to access e-government services. The elderly have often faced barriers due to limited digital capacity and lower levels of digital information proficiency compared with the general population. For example, in 2020, the digital information level among the elderly was reported to be 68.6, substantially lower than the national average of 100, clearly illustrating the digital divide (Ministry of Science and ICT 2021).

The COVID-19 pandemic has expedited the need for digital literacy and underscored the critical importance of bridging digital divide (Park and Park 2023). Simultaneously, the pandemic-induced changes in media and communication landscapes have

influenced the social capital of older adults and shaped their perceptions of welfare technology, emphasizing the urgent requirement for more inclusive digital policies (Kim, Park, Seo, Lim and Ahn 2022; Pyun 2022). Consequently, the pandemic has not only fundamentally restructured public service delivery but has also highlighted the pressing necessity of developing targeted digital inclusion strategies for the aging population. Because of the increased access to the e-government services, the research model is expected to show the degree of digital divide for the elderly during the pandemic and the practical validity, beyond the predicted "intention of use" in the Bélanger and Carter's model. We categorized the research respondents at three separate levels: (1) "no use" for the respondents who are informed of the e-government COVID-19 services but have not used them at all, (2) "partial use" for those who have used some of the services and (3) "use" for those who have used all the services.

III. Research Method

1. Research Model and Variables

This study proposes a modified research model based on the Bélanger and Carter's (2009) framework and investigates the determinants of the digital divide among the elderly. Our model assumes that determinants in the two main groups, 'access' and 'digital Capacity', are the main independent variables affecting the dependent variable, 'use of COVID-19 e-government services.' The model posits independent variables in two dimensions. The first dimension, 'access,' includes five demographic factors: household type, income, education, age and gender. The second dimension, 'digital Capacity,' includes the user's functional proficiency with two variables, each of which measures proficiency in personal computers (PC) and mobile devices respectively. The dependent variable, 'use of COVID-19 e-government service', represents the level of digital inclusion. It is categorically measured to see the degree of user engagement in online public services with three levels: 'no use' (aware of the service but not using it), 'partial use' (utilizing some available services) and 'full use' (utilizing all available services).

2. Sample

A total of 428 respondents between the ages of 55 and 89 across the nation were randomly selected for the sample. More females than males responded to the survey.

<Table 2> Demographics (N=428)

		N(%)		N(%)
Gender	Male	196(45.79)	Income Level (Monthly)	~1M 8(1.87)
	Female	232(54.21)		1M~1.99M 22(5.14)
Household Type	With Partner	410(95.79)		2M~2.99M 65(15.19)
	Without Partner	18(4.21)		3M~3.99M 93(21.73)
Education Level	Elementary School & Below	14(3.27)		4M~4.99M 111(25.93)
	Junior High School	39(9.11)		5M~5.99M 88(20.56)
	High School	296(69.16)		6M~ 41(9.58)
	University & Above	79(18.46)		No Use 105(24.53)
Age (Mean: 66.75, Min:55, Max: 89)	Use Level of e-government		Partial Use	111(25.94)
			Use	212(49.53)

Most of the respondents live with more than one family member; only 4.21% live alone. More than 60% of the respondents earned more than 3 million won per

month. Approximately nine out of the ten respondents had completed high school and/or attended other higher education institutes.

3. Analysis

We tested each hypothesis statistically and checked its significance. It was performed as follows. For the hypotheses that contained categorical values or ordinal values, which were H 1 (household type), H 2 (income), H 3 (education) and H 5 (gender), we conducted a chi-square test and realized that H 2, H 3 and H 5 were supported. For the hypotheses whose values were measured on a continuous scale, which were H 4 (Age), H 6a (Digital Capacity-PC) and H 6b (Digital Capacity-Mobile), we did an ANOVA test and realized that all of them were supported.

<Table 3> Analysis Result

	Hypothesis	Hired Test	Statistics	p	Supported
H 1	Household Type	Chi-square	.18	.91	No
H 2	Income	Chi-square	28.90	.00	Yes
H 3	Education	Chi-square	16.81	.01	Yes
H 4	Age(over 55)	ANOVA	44.97 (F-value)	.00	Yes
H 5	Gender	Chi-square	8.01	.02	Yes
H 6a	Digital Capacity-Computer	ANOVA	47.01 (F-value)	.00	Yes
H 6b	Digital Capacity-Mobile	ANOVA	45.95 (F-value)	.00	Yes

4. Result

The results show that all the proposed demographic independent variables, excluding household type, are significant. Typically, income level, education level and age are positively related to the level of use of the COVID-19 e-government services. However, some of our results contradict the findings of previous studies. First, whereas Bélanger and Carter (2009) argue that discussing the effect of gender would be pointless, we observed disparity in the digital divide between genders. Second, whereas some previous studies have indicated that elderly people living with younger generations have higher access to the Internet and are behaviorally inclined to use technology, our results show that whether or not a senior is living with a family member does not affect digital divide for the seniors during the pandemic. Third, both digital capacity with PC and digital capacity with mobile devices are shown to significantly affect the dependent variable, 'use of COVID-19 e-government services.' In addition, the post-hoc analysis shows that the more familiar the seniors are with PC and mobile devices, the higher the level of use of the COVID-19 e-government services is.

IV. Conclusion

In the aftermath of COVID-19, the call for digital inclusion has been made for digitally marginalized, particularly the elderly. In this study, we investigated what could affect digital divide for the elderly undergoing the COVID-19 pandemic. The findings reveal that demographic factors such as age, gender, income and education, frequently identified and discussed in previous studies, are still valid factors that affect the digital divide. Through this investigation, we advanced the initial theoretical ground, the Bélanger and Carter's framework. To be specific, we included new factors such as household type, gender and digital capacity with mobile devices and tested

for their significance, respectively. Based on our new model, in addition, the findings demonstrates that the effect of household type is not statistically significant. This finding is against some findings in the prior literature that a young family partner helps lessen digital divide for the senior family members. This inconsistency may be attributed to the small number of senior respondents living alone utilized in this study: only 18 of the total 428. As the society has seen an increase in the number of one-person households over the age of 55, the effect of household type on the digital divide needs to be investigated further on a large scale.

To bridge digital divide, this study suggests policy implications that education to foster digital capacity for the seniors matters. The research results indicate that on the side of access education significantly facilitates the use level of the COVID 19 e-government services and also that on the side of digital capacity digital proficiency in both PC and mobile devices does so. Therefore, continual education to improve digital capacity is needed for the elderly. Overall, for some parts, digital divide for the elderly stems from demographic factors. More importantly, however, education on digital capacity can lessen digital divide. For instance, public-private partnerships involving regional community centers and non-governmental organizations can be instrumental in delivering hands-on support for the seniors to get over digital divide.

This study contributes to both academic debate and policy regarding digital divide for the elderly. Today the population is aging and at the same time our life is being digitized rapidly. For sure the issue of digital divide for the elderly will be getting more important. In dealing with such digital divide issues, our findings imply that traditional structural determinants, particularly age and income, continue to shape digital divide, yet they also reveal the rising significance of instructional interventions such as education. Although this study presents academic implications regarding digital divide and suggestions to mitigate it, it is performed with some limitations. First, the research model did not include control variables because it was developed based on the original model from a previous research. Future research may set up an improved research model by considering possible control variables or independent variables. This approach may allow for more interpretation of digital divide. Second, the effect of household type on digital divide for the seniors still remains over controversy. While some previous studies accepted the effect of household type, this study rejected it. It needs to be filled up by future research.

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